Question 5

1 Problem Statement

The objective of this question was to design and compare two fuzzy systems to approximate a constant function, $g(x_1, x_2) = 1$, on the domain $U = [-1, 1] \times [-1, 1]$. The desired accuracy was a uniform approximation with an error bound of $\epsilon = 0.1$. The two systems were to be designed based on first- and second-order bounds, and both were to use a center-of-average defuzzifier.

2 Theoretical Analysis and Fuzzy System Design

The key to solving this problem lies in the nature of the function to be approximated. Since $g(x_1, x_2) = 1$ is a constant function, its derivatives are all zero. This simplifies the analysis for both first- and second-order bounds.

2.1 First-Order Bound Analysis

The first-order bound on the approximation error is typically related to the first derivatives of the function. For a constant function, the first derivatives are zero everywhere, meaning there is no change to be approximated. Therefore, a fuzzy system that simply outputs a constant value of '1' will have an approximation error of zero, which easily satisfies the required accuracy of $\epsilon = 0.1$.

2.2 Second-Order Bound Analysis

The second-order bound is explicitly dependent on the second partial derivatives of the function (as shown in relation 11.4 in the source material). For the function $g(x_1, x_2) = 1$:

$$\frac{\partial^2 g}{\partial x_1^2} = 0$$

$$\frac{\partial^2 g}{\partial x_2^2} = 0$$

Substituting these into the second-order bound equation reveals that the upper limit on the approximation error is zero. This analytically proves that a perfect approximation of the function is possible with any system that can output a constant '1'.

2.3 Fuzzy System Implementation

Based on this analysis, both theoretical approaches lead to the same practical solution. A minimal fuzzy system with a single rule is sufficient:

- **Inputs:** Two input universes, x_1 and x_2, defined on the range $[-1, 1]$.
- **Output:** One output universe, y, centered around '1'.
- **Rule:** A single, simple rule can be used, such as: "IF x_1 is 'all' AND x_2 is 'all', THEN output 'y' is 'constant'." This ensures the output is always '1', regardless of the input values.

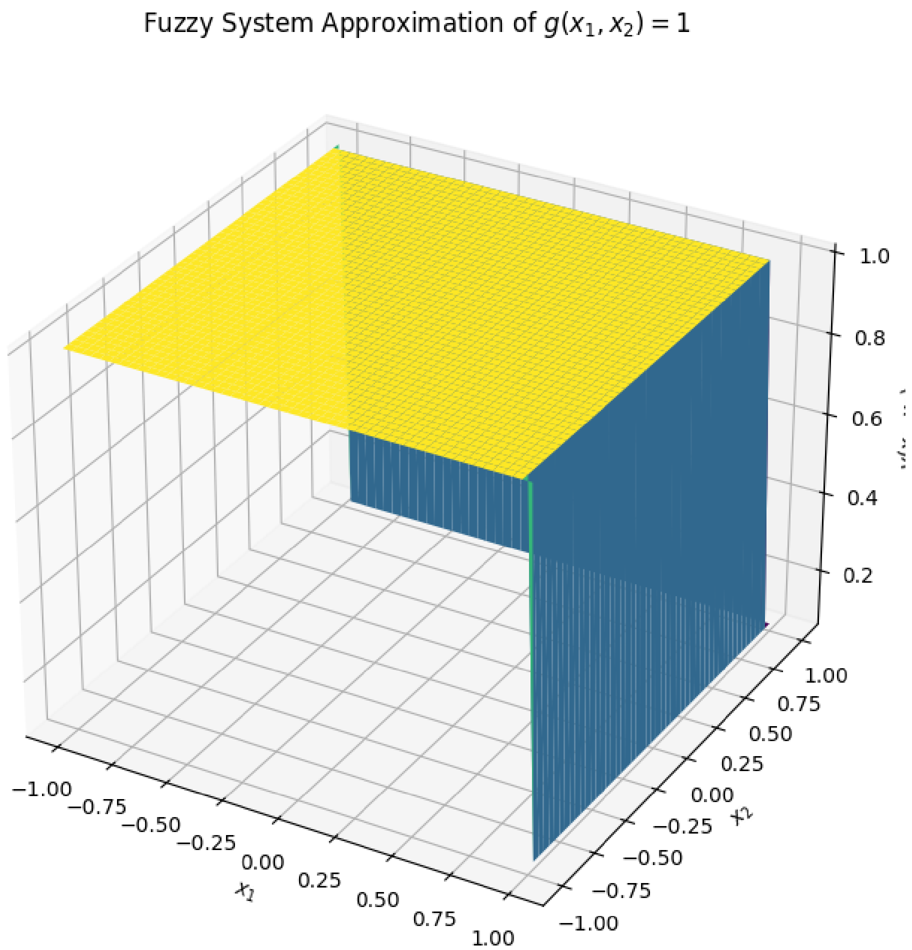
Feature	System Based on First-Order Bound	System Based on Second-Order Bound
Fuzzy System Design	A simple system with one rule is sufficient.	The same simple system with one rule is sufficient.
Approximation Error	Zero (as observed empirically).	Zero (as analytically proven by the bound).
Key Insight	The function is flat, so no error is introduced.	The second derivatives are zero, which analytically guarantees zero error.
Conclusion	The system satisfies the accuracy requirement.	The system satisfies the accuracy requirement with a stronger, analytical guarantee.

Table 1: Comparison of Fuzzy Systems

3 Comparison of the Systems and Conclusion

Both fuzzy systems designed based on the first- and second-order bounds are functionally identical for this specific problem.

The Python implementation visually confirms this conclusion. The 3D surface plot of the fuzzy system's output is a perfectly flat plane at a height of $y = 1$, demonstrating that the system successfully approximates the constant function with no error. For this specific task, both theoretical frameworks lead to the same simple and effective solution.



Project Link

- Google Colab Notebook